UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

SERIES 1 • CARD 08 / 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

CIRCULAR COORDINATES & DYNAMIC PROJECTIONS

Trigonometry is circular motion, not triangles.

On unit circle (r = 1), coordinates are trig values:

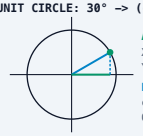
(X, Y) = (cos θ, sin θ)

Rotating disc projects angle θ onto cos/sin!

CALIBRATED 84.00 MM RADIUS DATUM

- Outer Ring: 84.00 mm ⌀ (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°

UNIT CIRCLE: 30° –> (cos = √3/2 ≈ 0.866, sin



ANGLE θ = 30° (π/6)

X = cos(30°) = √3/2 ≈ 0.8

Y = sin(30°) = 1/2 = 0.50

Pythagorean Identity:

cos²(θ) + sin²(θ) = 1

0.75 + 0.25 = 1.000

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
Unit Circle Disc	1x	84.0 mm ⌀	PLA White
Radian Overlay	1x	Radial Cursor	Clear PETG
Projection Arm	1x	Windows	Clear PC
Center Pivot	1x	Brass Bushing	Solid Brass
Stand & Lid	1x	Desk Stand	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts

QUADRANT SIGN CONVENTION

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (-, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (-, -): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, -): Cosine positive, Sine negative.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Align transparent cursor to 0° / 360°, slip the volvelle into its protective sleeve, and lay flat into the box recess. Snap the stand lid shut for silent, rattle-free carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° -> (cos 30° = √3/2, sin 30° = 1/2)

SERIES 1 • CARD 08 OF 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/unit-circle-volvelle/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS HSF.TF.A.2 (Unit Circle)
- CCSS HSF.TF.A.3 (Special Tria...)
- TEKS 111.42.c.2.P (Trig Functio...)

Grades 9–12 Trigonometry & Physics

GUIDING INQUIRY

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)

Turn the transparent cursor arm to the 30° mark.

Look at the inner radian scale: 30° aligns exactly with π/6 rad.

Notice the right triangle projected onto the unit circle base!

2 STEP 2: READ HORIZONTAL COSINE PROJECTION

Look down at the horizontal X-axis window:

X = cos(30°) = √3 / 2 ≈ 0.8660.

This is the horizontal base length of the 30°-60°-90° triangle.

3-4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC

3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.

4. VERIFY PYTHAGOREAN IDENTITY: cos²(30°) + sin²(30°) = 0.75 + 0.25 = 1.0000!

Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [⁻¹][•,x]) and run 45 [COS] [⁻¹] [x²] 45 [SIN] [⁻¹] [x²] [+] → 1.

5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)

Rotate cursor to 60°. Notice X and Y coordinates swap: X = cos(60°) = 1/2 = 0.500, Y = sin(60°) = √3/2 ≈ 0.866! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

VOLV-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:

Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 6 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California